

Soil profiles

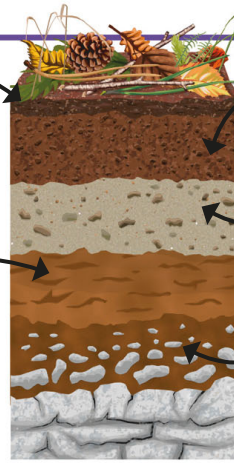
Soil is made up of broken rocks, minerals, decaying plants and animals, tiny creatures, gases, and water. If a section is cut through soil, you will see many layers. The depth of the layers vary in different soils.

Humus

Dead plants and animals decay, adding nutrients, or goodness, to the soil.

Subsoil

This moist layer contains clay and minerals from the weathered rock.



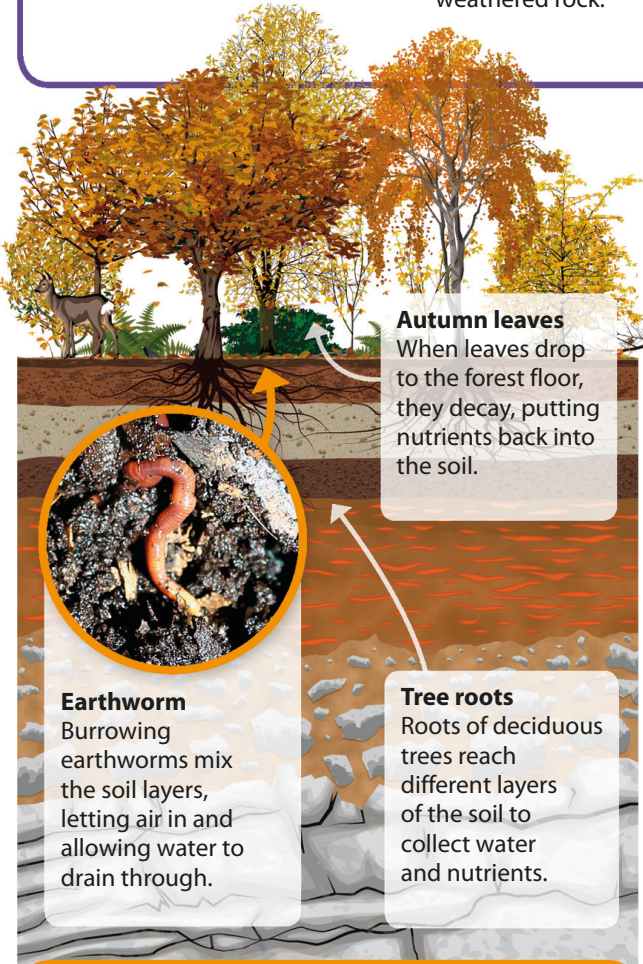
Top soil This dark layer is full of tiny creatures and nutrients, which plants use to grow.

Leaching layer

Water drips through this layer, leaving mostly sand and silt.

Weathered rock

Large rocks are broken up from the solid bedrock below.



Autumn leaves

When leaves drop to the forest floor, they decay, putting nutrients back into the soil.

Earthworm

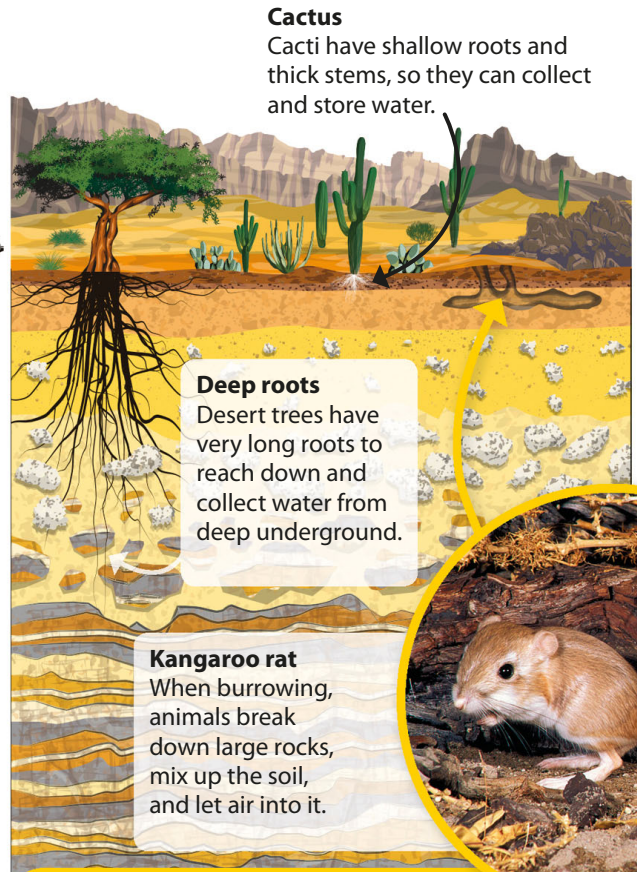
Burrowing earthworms mix the soil layers, letting air in and allowing water to drain through.

Tree roots

Roots of deciduous trees reach different layers of the soil to collect water and nutrients.

Temperate deciduous forest

The soil is moist and full of nutrients from decaying plants and animals, especially in autumn. The deep roots of plants break up the bedrock, which adds minerals. Water often drains through steadily.



Cactus

Cacti have shallow roots and thick stems, so they can collect and store water.

Deep roots

Desert trees have very long roots to reach down and collect water from deep underground.

Kangaroo rat

When burrowing, animals break down large rocks, mix up the soil, and let air into it.

Desert

The soil is very dry and few plants grow, so there is little humus. It is made up of boulders, pebbles, and sand, and is blown around by the wind. Any water drains through the sand easily.